```
> restart: unprotect('D'):
m := 1;
twist := + 0;
d := 8;                degre := d + 1:
##
HUU := X^(d+1) + T^(d+1);
#HUU := X^(d+1) + T^(d+1) + 1*X^floor((d+1)/2)*T^(d+1-floor((d+1)
/2));
##
UU  := subs({T=1,X=x}, HUU);
##
Jetyx := expand(
+ (1/Rz^(m-0*2)) * add(AA||j * x1^(m-0*3-j) * (y1)^j *
Delta_yx^0, j=0..m-0*3)
+ (1/Rz^(m-1*2)) * add(BB||j * x1^(m-1*3-j) * (y1)^j *
Delta_yx^1, j=0..m-1*3)
+ (1/Rz^(m-2*2)) * add(CC||j * x1^(m-2*3-j) * (y1)^j *
Delta_yx^2, j=0..m-2*3)
+ (1/Rz^(m-3*2)) * add(DD||j * x1^(m-3*3-j) * (y1)^j *
Delta_yx^3, j=0..m-3*3)
) :
##
dA := (m-0*2)*d - m + twist:
dB := (m-1*2)*d - m + twist:
dC := (m-2*2)*d - m + twist:
dD := (m-3*2)*d - m + twist:
##
Delta_yx := y1*x2 - y2*x1:
Soly1 := - x1*Rx/Ry - z1*Rz/Ry:
```

```
Soly2 := -x1^2*Rxx/Ry+2*x1^2*Rxy*Rx/Ry^2+2*x1*Rxy*z1*Rz/Ry^2-2*
x1*z1*Rxz/Ry-Ryy*x1^2*Rx^2/Ry^3-2*Ryy*x1*Rx*z1*Rz/Ry^3-Ryy*z1^2*
Rz^2/Ry^3+2*z1*Ryz*x1*Rx/Ry^2+2*z1^2*Ryz*Rz/Ry^2-z1^2*Rzz/Ry-x2*
Rx/Ry-z2*Rz/Ry:

##

Jetzx := sort(mtaylor(

                        expand(subs(x2=(Deltazx+x1*z2)/z1,
                         expand(    subs({y1=Soly1,y2=Soly2},
Jetyx)))),

         [x1,z1,Deltazx], 100), [x1,z1,Deltazx]):

##

printlevel := 2:
for k from 0 to m/3 do
for j from 0 to m-3*k do

EEEq[m-j-3*k,j,k] := factor(

 Rz^(m-j-3*k)
*
 Ry^(m-2*k)
*
  coeftayl(Jetzx, [x1,z1,Deltazx]=[0,0,0], [m-j-3*k,j,k])):

od: od:

## EQUATION RR EXPLICITE

RR := z^(d+1) + y^(d+1) + UU;

Rx := diff(RR,x):
Ry := diff(RR,y):
Rz := diff(RR,z):

Rxx := diff(RR,x,x):
Rxy := diff(RR,x,y):
Rxz := diff(RR,x,z):
Ryy := diff(RR,y,y):
Ryz := diff(RR,y,z):
Rzz := diff(RR,z,z):

##

printlevel := 2:
for k from 0 to m/3   do
for j from 0 to m-3*k do

EEEq[m-j-3*k,j,k] := mtaylor(EEEq[m-j-3*k,j,k], z, (m-j-3*k)*d):

od: od:

##
```

```
for j1 from 0 to 1+m*d+d do

ny[j1] := (-z^(d+1)-UU)^(floor(j1/(d+1))) * y^(j1-(d+1)*floor(j1/
(d+1))):

od:
```

$$m := 1$$
$$twist := 0$$
$$d := 8$$
$$HUU := T^9 + X^9$$
$$UU := x^9 + 1$$
$$RR := x^9 + y^9 + z^9 + 1 \quad (1)$$

```
> ##

for l from 0 to m-0*3 do  AA||l := add(A||l[j]*y^j, j=0..min(d,
dA)):  od:

for l from 0 to m-1*3 do  BB||l := add(B||l[j]*y^j, j=0..min(d,
dB)):  od:

for l from 0 to m-2*3 do  CC||l := add(C||l[j]*y^j, j=0..min(d,
dC)):  od:

for l from 0 to m-3*3 do  DD||l := add(D||l[j]*y^j, j=0..min(d,
dD)):  od:

##

1 + m*d + d:

printlevel := 2:
for k from 0 to m/3   do
for j from 0 to m-3*k do

degree(EEEq[m-j-3*k,j,k], y):

EEEq[m-j-3*k,j,k] := mtaylor(EEEq[m-j-3*k,j,k], y, 1+m*d+d):

dy[m-j-3*k,j,k]   :=  degree(EEEq[m-j-3*k,j,k], y):

od: od:

##

printlevel := 3:
for k from 0 to m/3   do
```

```
for j from 0 to m-3*k do

for j1 from 0 to dy[m-j-3*k,j,k] do

Cy[m-j-3*k,j,k,j1] := coeff(EEEq[m-j-3*k,j,k], y, j1):

od:

od: od:

##

printlevel := 2:
for k from 0 to m/3  do
for j from 0 to m-3*k do

EEq[m-j-3*k,j,k] := add(Cy[m-j-3*k,j,k,j1]*ny[j1], j1=0..dy[m-
j-3*k,j,k]):

degree(EEq[m-j-3*k,j,k], y):

od: od:

##

printlevel := 2:
for l from 0 to m-0*3    do
for j from 0 to min(d,dA) do

A||l[j] := add(A||l[j,k]*z^k, k=0..dA-j):

od: od:


printlevel := 2:
for l from 0 to m-1*3    do
for j from 0 to min(d,dB) do

B||l[j] := add(B||l[j,k]*z^k, k=0..dB-j):

od: od:


printlevel := 2:
for l from 0 to m-2*3    do
for j from 0 to min(d,dC) do

C||l[j] := add(C||l[j,k]*z^k, k=0..dC-j):

od: od:


printlevel := 2:
for l from 0 to m-3*3    do
for j from 0 to min(d,dD) do

D||l[j] := add(D||l[j,k]*z^k, k=0..dD-j):
```

```
od: od:

##

printlevel := 2:
for k from 0 to m/3   do
for j from 0 to m-3*k do

EEq[m-j-3*k,j,k] := mtaylor(EEq[m-j-3*k,j,k], z, (m-j-3*k)*d):

degree(EEq[m-j-3*k,j,k], z):

od: od:

##

printlevel := 3:
for l from 0 to m-0*3     do
for j from 0 to min(d,dA) do
for k from 0 to dA-j      do

A||l[j,k] := add(A||l[i,j,k]*x^i, i=0..dA-j-k):

od: od: od:


printlevel := 3:
for l from 0 to m-1*3     do
for j from 0 to min(d,dB) do
for k from 0 to dB-j      do

B||l[j,k] := add(B||l[i,j,k]*x^i, i=0..dB-j-k):

od: od: od:


printlevel := 3:
for l from 0 to m-2*3     do
for j from 0 to min(d,dC) do
for k from 0 to dC-j      do

C||l[j,k] := add(C||l[i,j,k]*x^i, i=0..dC-j-k):

od: od: od:


printlevel := 3:
for l from 0 to m-3*3     do
for j from 0 to min(d,dD) do
for k from 0 to dD-j      do

D||l[j,k] := add(D||l[i,j,k]*x^i, i=0..dD-j-k):

od: od: od:

##
```

```
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;

##

printlevel := 2:
for k from 0 to m/3  do
for j from 0 to m-3*k do

Eq[m-j-3*k,j,k] := mtaylor(EEq[m-j-3*k,j,k], z, (m-j-3*k)*d):

degree(Eq[m-j-3*k,j,k], z):

od: od:

##

printlevel := 2:
for k from 0 to m/3  do
for j from 0 to m-3*k do

degree(Eq[m-j-3*k,j,k], y):

od: od;
```

$$l := 0$$
$$\mathbf{120}$$
$$l := 1$$
$$\mathbf{120}$$
$$l := 2$$
$$l := 0$$
$$l := 0$$
$$l := 0$$
$$k := 0$$
$$\mathbf{8}$$
$$-\infty$$
$$k := 1 \tag{2}$$

```
> ## SYSTEM FOR DIVISIBILITY BY z^(1*d)

Systemez1d := {}:

for k from 0 to (m-1)/3 do

Coeffs_z1d[1,m-1-3*k,k] := {coeffs(expand(mtaylor(Eq[1,m-1-3*k,
k], z, 1*d)), [x,y,z])}:

Systemez1d := Systemez1d union Coeffs_z1d[1,m-1-3*k,k]:

od:

Systeme_z1d := Systemez1d:

## VARIABLES FOR DIVISIBILITY BY z^(1*d)

Variables_z1d := indets(Systeme_z1d):

## RESOLUTION z^(1*d)

Solutions_z1d := solve(Systeme_z1d, Variables_z1d):

assign(Solutions_z1d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$l := 0$$
$$l := 0$$
$$l := 0 \quad (3)$$

```
> ## SYSTEM FOR DIVISIBILITY BY z^(2*d)

Systemez2d := {}:

for k from 0 to (m-2)/3 do
```

```
Coeffs_z2d[2,m-2-3*k,k] := {coeffs(expand(mtaylor(Eq[2,m-2-3*k,
k], z, 2*d)), [x,y,z])}:

Systemez2d := Systemez2d union Coeffs_z2d[2,m-2-3*k,k]:

od:

Systeme_z2d := Systemez2d:

## VARIABLES FOR DIVISIBILITY BY z^(2*d)

Variables_z2d := indets(Systeme_z2d):

## RESOLUTION z^(2*d)

Solutions_z2d := solve(Systeme_z2d, Variables_z2d):

assign(Solutions_z2d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{4}$$

```
> ## SYSTEM FOR DIVISIBILITY BY z^(3*d)

Systemez3d := {}:

for k from 0 to (m-3)/3 do

Coeffs_z3d[3,m-3-3*k,k] := {coeffs(expand(mtaylor(Eq[3,m-3-3*k,
k], z, 3*d)), [x,y,z])}:

Systemez3d := Systemez3d union Coeffs_z3d[3,m-3-3*k,k]:

od:

Systeme_z3d := Systemez3d:
```

```
## VARIABLES FOR DIVISIBILITY BY z^(3*d)

Variables_z3d := indets(Systeme_z3d):

## RESOLUTION z^(3*d)

Solutions_z3d := solve(Systeme_z3d, Variables_z3d):

assign(Solutions_z3d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{5}$$

```
> ## SYSTEM FOR DIVISIBILITY BY z^(4*d)

Systemez4d := {}:

for k from 0 to (m-4)/3 do

Coeffs_z4d[4,m-4-3*k,k] := {coeffs(expand(mtaylor(Eq[4,m-4-3*k,
k], z, 4*d)), [x,y,z])}:

Systemez4d := Systemez4d union Coeffs_z4d[4,m-4-3*k,k]:

od:

Systeme_z4d := Systemez4d:

## VARIABLES FOR DIVISIBILITY BY z^(4*d)

Variables_z4d := indets(Systeme_z4d):

## RESOLUTION z^(4*d)

Solutions_z4d := solve(Systeme_z4d, Variables_z4d):
```

```
assign(Solutions_z4d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$l := 0$$
$$l := 0$$
$$l := 0 \qquad (6)$$

```
> ## SYSTEM FOR DIVISIBILITY BY z^(5*d)

Systemez5d := {}:

for k from 0 to (m-5)/3 do

Coeffs_z5d[5,m-5-3*k,k] := {coeffs(expand(mtaylor(Eq[5,m-5-3*k,
k], z, 5*d)), [x,y,z])}:

Systemez5d := Systemez5d union Coeffs_z5d[5,m-5-3*k,k]:

od:

Systeme_z5d := Systemez5d:

## VARIABLES FOR DIVISIBILITY BY z^(5*d)

Variables_z5d := indets(Systeme_z5d):

## RESOLUTION z^(5*d)

Solutions_z5d := solve(Systeme_z5d, Variables_z5d):

assign(Solutions_z5d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{7}$$

```
> ## SYSTEM FOR DIVISIBILITY BY z^(6*d)
Systemez6d := {}:
for k from 0 to (m-6)/3 do
Coeffs_z6d[6,m-6-3*k,k] := {coeffs(expand(mtaylor(Eq[6,m-6-3*k,
k], z, 6*d)), [x,y,z])}:
Systemez6d := Systemez6d union Coeffs_z6d[6,m-6-3*k,k]:
od:
Systeme_z6d := Systemez6d:
## VARIABLES FOR DIVISIBILITY BY z^(6*d)
Variables_z6d := indets(Systeme_z6d):
## RESOLUTION z^(6*d)
Solutions_z6d := solve(Systeme_z6d, Variables_z6d):
assign(Solutions_z6d):
##
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$l := 0$$

$l := 0$

$l := 0$ (8)

```
> ## SYSTEM FOR DIVISIBILITY BY z^(7*d)
Systemez7d := {}:
for k from 0 to (m-7)/3 do
Coeffs_z7d[7,m-7-3*k,k] := {coeffs(expand(mtaylor(Eq[7,m-7-3*k,
k], z, 7*d)), [x,y,z])}:
Systemez7d := Systemez7d union Coeffs_z7d[7,m-7-3*k,k]:
od:
Systeme_z7d := Systemez7d:
## VARIABLES FOR DIVISIBILITY BY z^(7*d)
Variables_z7d := indets(Systeme_z7d):
## RESOLUTION z^(7*d)
Solutions_z7d := solve(Systeme_z7d, Variables_z7d):
assign(Solutions_z7d):
##
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

$l := 0$

1

$l := 1$

1

$l := 2$

$l := 0$

$l := 0$

$l := 0$ (9)

```
> ## SYSTEM FOR DIVISIBILITY BY z^(8*d)
```

```
Systemez8d := {}:

for k from 0 to (m-8)/3 do

Coeffs_z8d[8,m-8-3*k,k] := {coeffs(expand(mtaylor(Eq[8,m-8-3*k,
k], z, 8*d)), [x,y,z])}:

Systemez8d := Systemez8d union Coeffs_z8d[8,m-8-3*k,k]:

od:

Systeme_z8d := Systemez8d:

## VARIABLES FOR DIVISIBILITY BY z^(8*d)

Variables_z8d := indets(Systeme_z8d):

## RESOLUTION z^(8*d)

Solutions_z8d := solve(Systeme_z8d, Variables_z8d):

assign(Solutions_z8d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$l := 0$$
$$l := 0$$
$$l := 0 \qquad \textbf{(10)}$$

```
> ## SYSTEM FOR DIVISIBILITY BY z^(9*d)

Systemez9d := {}:

for k from 0 to (m-9)/3 do

Coeffs_z9d[9,m-9-3*k,k] := {coeffs(expand(mtaylor(Eq[9,m-9-3*k,
k], z, 9*d)), [x,y,z])}:

Systemez9d := Systemez9d union Coeffs_z9d[9,m-9-3*k,k]:
```

```
od:
Systeme_z9d := Systemez9d:
## VARIABLES FOR DIVISIBILITY BY z^(9*d)
Variables_z9d := indets(Systeme_z9d):
## RESOLUTION z^(9*d)
Solutions_z9d := solve(Systeme_z9d, Variables_z9d):
assign(Solutions_z9d):
##
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{11}$$

```
> ## SYSTEM FOR DIVISIBILITY BY z^(10*d)
Systemez10d := {}:
for k from 0 to (m-10)/3 do
Coeffs_z10d[10,m-10-3*k,k] := {coeffs(expand(mtaylor(Eq[10,m-10
-3*k,k], z, 10*d)), [x,y,z])}:
Systemez10d := Systemez10d union Coeffs_z10d[10,m-10-3*k,k]:
od:
Systeme_z10d := Systemez10d:
## VARIABLES FOR DIVISIBILITY BY z^(10*d)
Variables_z10d := indets(Systeme_z10d):
## RESOLUTION z^(10*d)
```

```
Solutions_z10d := solve(Systeme_z10d, Variables_z10d):

assign(Solutions_z10d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{12}$$

```
> ## SYSTEM FOR DIVISIBILITY BY z^(11*d)

Systemez11d := {}:

for k from 0 to (m-11)/3 do

Coeffs_z11d[11,m-11-3*k,k] := {coeffs(expand(mtaylor(Eq[11,m-11
-3*k,k], z, 11*d)), [x,y,z])}:

Systemez11d := Systemez11d union Coeffs_z11d[11,m-11-3*k,k]:

od:

Systeme_z11d := Systemez11d:

## VARIABLES FOR DIVISIBILITY BY z^(11*d)

Variables_z11d := indets(Systeme_z11d):

## RESOLUTION z^(11*d)

Solutions_z11d := solve(Systeme_z11d, Variables_z11d):

assign(Solutions_z11d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
```

```
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

$$l := 0$$

$$\mathbf{1}$$

$$l := 1$$

$$\mathbf{1}$$

$$l := 2$$

$$l := 0$$

$$l := 0$$

$$l := 0 \tag{13}$$

```
> ## HOLOMORPHICITY AT INFINITY

for l from 0 to m - 0*3 do

HA||l := add(add(add(A||l[i,j,k]*X^i*Z^k*Y^j*T^(dA-i-j-k), i=0..
dA-j-k), k=0..dA-j), j=0..min(d,dA)):

od:

#

for l from 0 to m - 1*3 do

HB||l := add(add(add(B||l[i,j,k]*X^i*Z^k*Y^j*T^(dB-i-j-k), i=0..
dB-j-k), k=0..dB-j), j=0..min(d,dB)):

od:

#

for l from 0 to m - 2*3 do

HC||l := add(add(add(C||l[i,j,k]*X^i*Z^k*Y^j*T^(dC-i-j-k), i=0..
dC-j-k), k=0..dC-j), j=0..min(d,dC)):

od:

#
```

```
for l from 0 to m - 3*3 do

HD||l := add(add(add(D||l[i,j,k]*X^i*Z^k*Y^j*T^(dD-i-j-k), i=0..
dD-j-k), k=0..dD-j), j=0..min(d,dD)):

od:

##

for h from 0 to m - 3*0 do

HAEEq[h] := add(HA||l*binomial(l,h)*X^(m-3*0-l)*Y^(l-h), l=h..
m-3*0):
dYA[h]  := degree(HAEEq[h], Y):

od:

#

for h from 0 to m - 3*1 do

HBEEq[h] := add(HB||l*binomial(l,h)*X^(m-3*1-l)*Y^(l-h), l=h..
m-3*1):
dYB[h]  := degree(HBEEq[h], Y):

od:

#

for h from 0 to m - 3*2 do

HCEEq[h] := add(HC||l*binomial(l,h)*X^(m-3*2-l)*Y^(l-h), l=h..
m-3*2):
dYC[h]  := degree(HCEEq[h], Y):

od:

#

for h from 0 to m - 3*3 do

HDEEq[h] := add(HD||l*binomial(l,h)*X^(m-3*3-l)*Y^(l-h), l=h..
m-3*3):
dYD[h]  := degree(HDEEq[h], Y):

od:

##

for h  from 0 to m - 3*0 do
for j1 from 0 to dYA[h]  do

CoeffA[h,j1] := coeff(HAEEq[h], Y, j1):

od: od:

#
```

```
for h  from 0 to m - 3*1 do
for j1 from 0 to dYB[h]  do

CoeffB[h,j1] := coeff(HBEEq[h], Y, j1):

od: od:

#

for h  from 0 to m - 3*2 do
for j1 from 0 to dYC[h]  do

CoeffC[h,j1] := coeff(HCEEq[h], Y, j1):

od: od:

#

for h  from 0 to m - 3*3 do
for j1 from 0 to dYD[h]  do

CoeffD[h,j1] := coeff(HDEEq[h], Y, j1):

od: od:

##

for j1 from 0 to 1+m*d+d do

nY[j1] := (-Z^(d+1)-HUU)^(floor(j1/(d+1))) * Y^(j1-(d+1)*floor
(j1/(d+1))):

od:

##

for h from 0 to m - 3*0 do

HAEq[h] := expand(mtaylor(add(CoeffA[h,j1]*nY[j1], j1=0..dYA[h]),
T, m-3*0-h)):
degree(HAEq[h],Y):

od:

#

for h from 0 to m - 3*1 do

HBEq[h] := expand(mtaylor(add(CoeffB[h,j1]*nY[j1], j1=0..dYB[h]),
T, m-3*1-h)):
degree(HBEq[h],Y):

od:

#

for h from 0 to m - 3*2 do
```

```
HCEq[h] := expand(mtaylor(add(CoeffC[h,j1]*nY[j1], j1=0..dYC[h]),
T, m-3*2-h)):
degree(HCEq[h],Y):

od:

#

for h from 0 to m - 3*3 do

HDEq[h] := expand(mtaylor(add(CoeffD[h,j1]*nY[j1], j1=0..dYD[h]),
T, m-3*3-h)):
degree(HDEq[h],Y):

od:
```

```
> HSysteme :=      {seq(coeffs(HAEq[h],[T,X,Y,Z]),h=0..m-3*0)}
            union {seq(coeffs(HBEq[h],[T,X,Y,Z]),h=0..m-3*1)}
            union {seq(coeffs(HCEq[h],[T,X,Y,Z]),h=0..m-3*2)}
            union {seq(coeffs(HDEq[h],[T,X,Y,Z]),h=0..m-3*3)}
minus {0}:

HSolutions := solve(HSysteme):

assign(HSolutions):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{14}$$

```
> Jetyx := factor(

  + (1/Rz^(m-0*2)) * x^(-twist) * add(AA||j * x1^(m-0*3-j) * (y1)^j
  * Delta_yx^0, j=0..m-0*3)

  + (1/Rz^(m-1*2)) * x^(-twist) * add(BB||j * x1^(m-1*3-j) * (y1)^j
  * Delta_yx^1, j=0..m-1*3)

  + (1/Rz^(m-2*2)) * x^(-twist) * add(CC||j * x1^(m-2*3-j) * (y1)^j
  * Delta_yx^1, j=0..m-2*3)

  + (1/Rz^(m-3*2)) * x^(-twist) * add(DD||j * x1^(m-3*3-j) * (y1)^j
  * Delta_yx^1, j=0..m-3*3)

  ) :
```

```
> Vars := indets(Jetyx) minus {x, x1, x2, y, y1, y2, z}:

  NbVars := nops(Vars);
```

(15)

$$NbVars := 0 \tag{15}$$

```
> Ens_Vars := {}:
for i from 1 to nops(Vars) do
Ens_Vars := Ens_Vars union {op(i,Vars)=0}:
od:
EnsVars := Ens_Vars:
```

```
> for i from 1 to nops(Vars) do
JetSol[i] := factor(subs(EnsVars, subs(op(i,Vars)=1, Jetyx))):
od:
```

```
> ## VERIICATION OF HOLOMORPHICITY AT INFINITY
Liste_x := {x = 1/x,
              y = y/x,
              z = z/x,
             x1 = - x1/x^2,
            x2 = 2*x1^2/x^3 - x2/x^2,
             y1 = y1/x - y*x1/x^2,
           y2 = y2/x - 2*y1*x1/x^2 + 2*y*x1^2/x^3 - y*x2/x^2}:
##
for i from 1 to nops(Vars) do
```

```
JetSol_x[i] := factor(subs(Liste_x, JetSol[i])):
od:
```

```
> Vars := indets(Jetyx) minus {x, x1, x2, y, y1, y2, z}:
NbVars := nops(Vars);
```

$$NbVars := 0 \tag{16}$$

```
> ## SEE AGAIN THE SOLUTIONS
for i from 1 to nops(Vars) do
JetSol[i] := factor(subs(EnsVars, subs(op(i,Vars)=1, Jetyx))):
od;
```

$$i := 1 \tag{17}$$